

HOT AND MIDLATITUDE DESERT CLIMATE

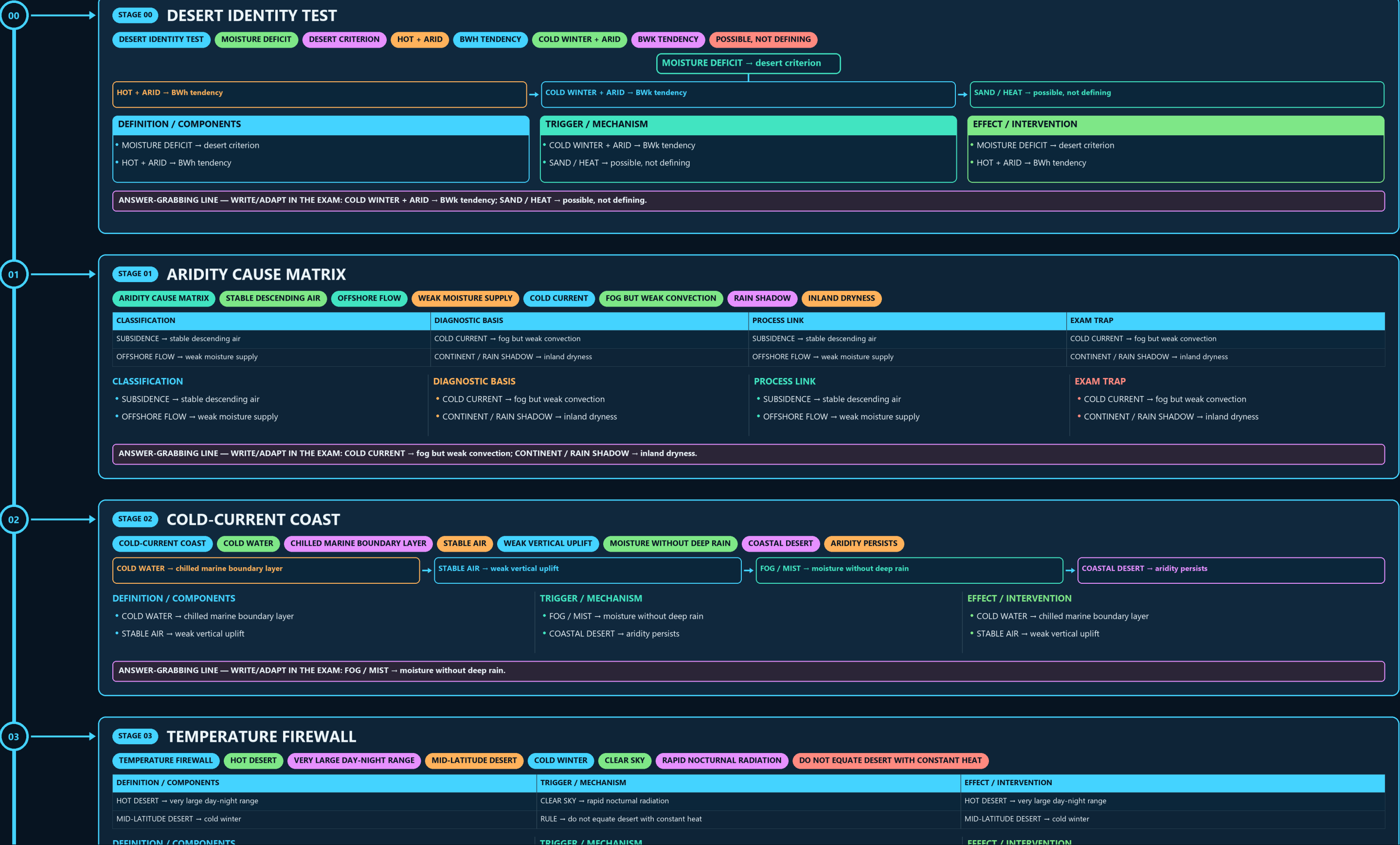
DESERT IDENTITY TEST → ARIDITY CAUSE MATRIX → COLD-CURRENT COAST → TEMPERATURE FIREWALL → WADI FLOOD CHAIN → PLANT-WATER ADAPTATIONS

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

PRIMARY CORE FLOW SUBORDINATE EXTRA APPROVAL / REVIEW

Approval: FALSE • Pending user review • active generation g2 • explicit approval of this exact generation is required • all prior artifacts unchanged

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles



- COLD WATER → chilled marine boundary layer
- STABLE AIR → weak vertical uplift

- FOG / MIST → moisture without deep rain
- COASTAL DESERT → aridity persists

- COLD WATER → chilled marine boundary layer
- STABLE AIR → weak vertical uplift

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: FOG / MIST → moisture without deep rain.

03

STAGE 03

TEMPERATURE FIREWALL

TEMPERATURE FIREWALLHOT DESERTVERY LARGE DAY-NIGHT RANGEMID-LATITUDE DESERTCOLD WINTERCLEAR SKYRAPID NOCTURNAL RADIATIONDO NOT EQUATE DESERT WITH CONSTANT HEAT

DEFINITION / COMPONENTS	TRIGGER / MECHANISM	EFFECT / INTERVENTION
HOT DESERT → very large day-night range	CLEAR SKY → rapid nocturnal radiation	HOT DESERT → very large day-night range
MID-LATITUDE DESERT → cold winter	RULE → do not equate desert with constant heat	MID-LATITUDE DESERT → cold winter

DEFINITION / COMPONENTS

- HOT DESERT → very large day-night range
- MID-LATITUDE DESERT → cold winter

TRIGGER / MECHANISM

- CLEAR SKY → rapid nocturnal radiation
- RULE → do not equate desert with constant heat

EFFECT / INTERVENTION

- HOT DESERT → very large day-night range
- MID-LATITUDE DESERT → cold winter

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → do not equate desert with constant heat.

04

STAGE 04

WADI FLOOD CHAIN

WADI FLOOD CHAINRARE INTENSE STORMHIGH RAIN RATECRUSTED SURFACERAPID RUNOFFDRY CHANNELSUDDEN FLOW CONCENTRATIONFLASH-FLOOD EXPOSURE

RARE INTENSE STORM → high rain rate → BARE / CRUSTED SURFACE → rapid runoff → DRY CHANNEL → sudden flow concentration → SETTLEMENT / ROAD → flash-flood exposure

DEFINITION / COMPONENTS	TRIGGER / MECHANISM	EFFECT / INTERVENTION
RARE INTENSE STORM → high rain rate	DRY CHANNEL → sudden flow concentration	RARE INTENSE STORM → high rain rate
BARE / CRUSTED SURFACE → rapid runoff	SETTLEMENT / ROAD → flash-flood exposure	BARE / CRUSTED SURFACE → rapid runoff

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: DRY CHANNEL → sudden flow concentration; SETTLEMENT / ROAD → flash-flood exposure.

05

STAGE 05

PLANT-WATER ADAPTATIONS

PLANT-WATER ADAPTATIONSREDUCED, WAXY OR THORNYWATER STORAGE IN SOME SPECIESDEEP TAP OR WIDE SHALLOW NETWORKLOWER COMPETITION FOR WATER

LEAF → reduced, waxy or thorny

STEM → water storage in some species → ROOT → deep tap or wide shallow network → SPACING → lower competition for water

CLASSIFICATION	DIAGNOSTIC BASIS	PROCESS LINK	EXAM TRAP
LEAF → reduced, waxy or thorny STEM → water storage in some species	ROOT → deep tap or wide shallow network SPACING → lower competition for water	LEAF → reduced, waxy or thorny STEM → water storage in some species	ROOT → deep tap or wide shallow network SPACING → lower competition for water

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: ROOT → deep tap or wide shallow network.

06

STAGE 06

OASIS AND WATER STRATEGY

OASIS AND WATER STRATEGYEXOTIC RIVERDATE PALM + CROPSLAYERED USEDEPLETION, SALINITY OR CONFLICT

AQUIFER / SPRING / EXOTIC RIVER → source

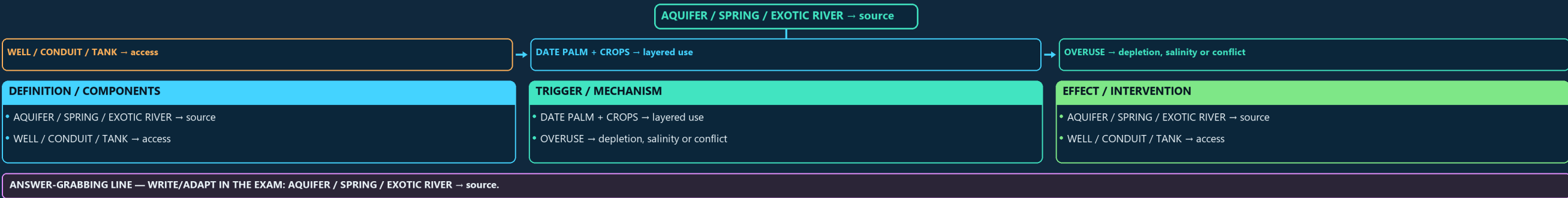
WELL / CONDUIT / TANK → access → DATE PALM + CROPS → layered use → OVERUSE → depletion, salinity or conflict

DEFINITION / COMPONENTS	TRIGGER / MECHANISM	EFFECT / INTERVENTION
AQUIFER / SPRING / EXOTIC RIVER → source WELL / CONDUIT / TANK → access	DATE PALM + CROPS → layered use OVERUSE → depletion, salinity or conflict	AQUIFER / SPRING / EXOTIC RIVER → source WELL / CONDUIT / TANK → access

06

STAGE 06 OASIS AND WATER STRATEGY

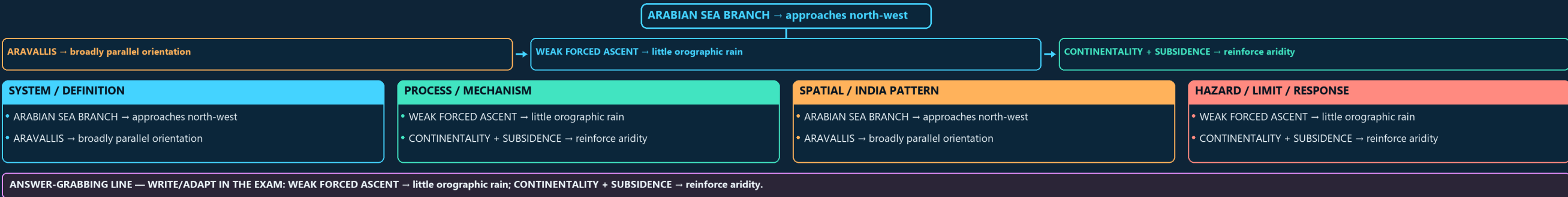
OASIS AND WATER STRATEGY EXOTIC RIVER DATE PALM + CROPS LAYERED USE DEPLETION, SALINITY OR CONFLICT



07

STAGE 07 THAR ARIDITY MAP

THAR ARIDITY MAP ARABIAN SEA BRANCH APPROACHES NORTH-WEST BROADLY PARALLEL ORIENTATION WEAK FORCED ASCENT LITTLE OROGRAPHIC RAIN CONTINENTALITY + SUBSIDENCE REINFORCE ARIDITY



08

STAGE 08 THAR LANDSCAPE LEDGER

THAR LANDSCAPE LEDGER BARCHAN, SEIF AND SAND SHEETS INLAND SALT BASINS PRINCIPAL INLAND-DRAINING RIVER XERIC OPEN HABITAT

DEFINITION / COMPONENTS	TRIGGER / MECHANISM	EFFECT / INTERVENTION
DUNES → barchan, seif and sand sheets	LUNI → principal inland-draining river	DUNES → barchan, seif and sand sheets
DHANDS / PLAYAS → inland salt basins	KHEJRI / GRASS → xeric open habitat	DHANDS / PLAYAS → inland salt basins
DEFINITION / COMPONENTS	TRIGGER / MECHANISM	EFFECT / INTERVENTION
- DUNES → barchan, seif and sand sheets - DHANDS / PLAYAS → inland salt basins	- LUNI → principal inland-draining river - KHEJRI / GRASS → xeric open habitat	- DUNES → barchan, seif and sand sheets - DHANDS / PLAYAS → inland salt basins

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: LUNI → principal inland-draining river; KHEJRI / GRASS → xeric open habitat.

09

STAGE 09 TRANSFORMATION TRADE-OFF

TRANSFORMATION TRADE-OFF LOCAL HARVESTING IRRIGATION AND SETTLEMENT EXPANSION RENEWABLE GRID LOW-CARBON POWER SALINITY, WATERLOGGING, HABITAT COLLISION

DEFINITION / COMPONENTS	TRIGGER / MECHANISM	EFFECT / INTERVENTION
TANKA / KHADIN / JOHAD → local harvesting	RENEWABLE GRID → low-carbon power	TANKA / KHADIN / JOHAD → local harvesting
CANAL → irrigation and settlement expansion	RISKS → salinity, waterlogging, habitat collision	CANAL → irrigation and settlement expansion
DEFINITION / COMPONENTS	TRIGGER / MECHANISM	EFFECT / INTERVENTION
- TANKA / KHADIN / JOHAD → local harvesting - CANAL → irrigation and settlement expansion	- RENEWABLE GRID → low-carbon power - RISKS → salinity, waterlogging, habitat collision	- TANKA / KHADIN / JOHAD → local harvesting - CANAL → irrigation and settlement expansion

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: TANKA / KHADIN / JOHAD → local harvesting

09

STAGE 09

TRANSFORMATION TRADE-OFF

TRANSFORMATION TRADE-OFFLOCAL HARVESTINGIRRIGATION AND SETTLEMENT EXPANSIONRENEWABLE GRIDLOW-CARBON POWERSALINITY, WATERLOGGING, HABITAT COLLISION

DEFINITION / COMPONENTS	TRIGGER / MECHANISM	EFFECT / INTERVENTION
TANKA / KHADIN / JOHAD → local harvesting	RENEWABLE GRID → low-carbon power	TANKA / KHADIN / JOHAD → local harvesting
CANAL → irrigation and settlement expansion	RISKS → salinity, waterlogging, habitat collision	CANAL → irrigation and settlement expansion

DEFINITION / COMPONENTS

- TANKA / KHADIN / JOHAD → local harvesting
- CANAL → irrigation and settlement expansion

TRIGGER / MECHANISM

- RENEWABLE GRID → low-carbon power
- RISKS → salinity, waterlogging, habitat collision

EFFECT / INTERVENTION

- TANKA / KHADIN / JOHAD → local harvesting
- CANAL → irrigation and settlement expansion

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: TANKA / KHADIN / JOHAD → local harvesting.

10

STAGE 10

UNCCD STATUS GATE

UNCCD STATUS GATERIYADH, 2-13 DECEMBER 2024LAND + DROUGHTRESILIENCE AGENDA ADVANCEDBINDING FRAMEWORKNOT FINALISDPYQ LEDGERNO DIRECT TOPIC 18 DEMAND

COP16 → Riyadh, 2-13 December 2024

LAND + DROUGHT → resilience agenda advanced

BINDING FRAMEWORK → not finalised

PYQ LEDGER → no direct Topic 18 demand

DEFINITION / COMPONENTS

- COP16 → Riyadh, 2-13 December 2024
- LAND + DROUGHT → resilience agenda advanced

TRIGGER / MECHANISM

- BINDING FRAMEWORK → not finalised
- PYQ LEDGER → no direct Topic 18 demand

EFFECT / INTERVENTION

- COP16 → Riyadh, 2-13 December 2024
- LAND + DROUGHT → resilience agenda advanced

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: PYQ LEDGER → no direct Topic 18 demand.

11

STAGE 11

DESERT ANSWER SPINE

DESERT ANSWER SPINEARIDITY AND WATER BALANCECIRCULATION, CURRENT AND RELIEF CAUSES

DEFINE → aridity and water balance

MAP → circulation, current and relief causes

COMPARE → hot versus mid-latitude

APPLY → Thar, GIB and drought governance

DEFINITION / COMPONENTS

- DEFINE → aridity and water balance
- MAP → circulation, current and relief causes

TRIGGER / MECHANISM

- COMPARE → hot versus mid-latitude
- APPLY → Thar, GIB and drought governance

EFFECT / INTERVENTION

- DEFINE → aridity and water balance
- MAP → circulation, current and relief causes

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: COMPARE → hot versus mid-latitude; APPLY → Thar, GIB and drought governance.

E

EXTRA

SUBORDINATE ENRICHMENT — ONLY AFTER THE COMPLETE CORE

OPTIONAL ADVANCED ENRICHMENTTHAR COREWESTERN RAJASTHAN, WEST OF ARAVALLIS; NORTHERN GUJARAT.THE SOURCE-ERA WESTERN-CORE EXAMPLE GIVES ABOUT 45°C10°C FOR JULY/JANUARY; STATION MEANS VARY.MEAN ANNUAL RAINFALL<25 CM .KEY ARIDITY CAUSE

OPTIONAL PROCESS DEPTH

- Thar core: western Rajasthan, west of Aravallis; northern Gujarat.
- The source-era western-core example gives about 45°C / 10°C for July/January; station means vary.
- Mean annual rainfall: <25 cm .
- Key aridity cause: Aravallis lie parallel to Arabian Sea monsoon current.

DATA / INSTITUTION LIMIT

- Landforms: barchan and seif dunes; playas/dhands; Sambhar salt lake.
- Luni is the main inland-draining river.
- Khejri = important desert tree / Rajasthan state tree in source notes.
- Bajra, pulses and fodder are characteristic dryland crops in the source text.

CONTEMPORARY PARALLEL

- Traditional water harvesting: tanka, khadin, johad.
- Indira Gandhi Nahar transformed parts of the desert.
- NOT: Aravallis block the monsoon and should make Thar wet → They run parallel to the Arabian Sea branch and fail to force uplift.
- NOT: Thar is uninhabited like a core desert → It is densely populated, with pastoralism, towns and canal irrigation.

OPTIONAL STATUS — This optional depth is unnecessary for a competent core answer; use it only for added nuance after the complete core has been written.